

Guardforce AI: Delivering robotic solutions to users around the world with Google Cloud

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ABOUT GUARDFORCE AI

Leveraging voice recognition technology, high-performance computing resources on Google Cloud, Guardforce AI is able to provide robotic solutions in less than 10 languages with approximately 60% less labor costs.



About Guardforce AI

Guardforce AI is a global security solution provider with more than 41 years of experience in the security and logistics industry. It tapped into the AI domain in 2020 and started offering AI and robotic solutions that can help enterprises enhance operational efficiency. As of now, Guardforce AI has formed 3 types of AI and Robotics solutions, providing functions such as hotel self-check-in/out, AIoT robot advertising, event management, security solutions, food delivery services, disinfection services, and more.

halving its product development cycle

Industries: Technology

Location: Singapore

Google Cloud results

- Supports a global technological framework for robotic solutions to save approximately 60 percent of labor costs
- Improves interactions between humans and robots in less spoken languages with speech-to-text and text-to-speech AI
- Helps ensure compliance with data privacy

Short product development cycle

(<https://cloud.google.com/media-cdn>), [Cloud](#)



About Master Concept

A Google Cloud partner, Master Concept was founded in 2003 and is committed to providing professional technology and cloud consulting services to world-class leading brands. Master Concept currently serves more than 3,000 enterprise customers in various industries in the Asia-Pacific region. Master Concept is listed as one of the most experienced cloud computing solution providers in the Asia-Pacific region.



Google Cloud (<https://cloud.google.com/>)

Cloud CDN (<https://cloud.google.com/cdn/>)

Media CDN (<https://cloud.google.com/media-cdn>)

Cloud Storage (<https://cloud.google.com/storage/>)

Compute Engine

(<https://cloud.google.com/compute/>)

Dialogflow (<https://cloud.google.com/dialogflow>)

serves more than 3,000 enterprise customers in various industries in the

regulations
across
regions to
enable quick
business
expansion

Cloud Speech to Text

(<https://cloud.google.com/speech/>)

cloud computing solution providers in the

Cloud Text to Speech

(<https://cloud.google.com/text-to-speech/>)

Guardforce AI (<https://www.guardforceai.com/>) is a Nasdaq-listed company founded in 1982 in Thailand as a security and logistics solution provider. For more than 40 years, it has been offering cash management and transport services to commercial banks and retailers in the Asian Pacific region. When COVID-19 brought about the trend of intelligentization across industries in 2020, Guardforce AI saw great application potentials of service robots and decided to start developing robotic solutions for its enterprise customers to help enhance their operational efficiency through automation.

Most of Guardforce AI's competitors in the robotics domain sell their hardware and software products together as packaged solutions. To set itself apart in the fierce competition in the robot manufacturing industry, Guardforce AI chose to develop a Robot-as-a-Service (RaaS) business model centered around its self-built intelligent cloud platform. Its customers can freely opt for buying or renting its service robots, which can be customized when needed.

Guardforce AI's Intelligent Cloud Platform (GFAI ICP) provides hardware, software interfaces and cloud APIs to enable its robots to connect and

communicate with various devices, software and platform services. This way, its robotic solutions can be made compatible with different systems and technological frameworks. Guardforce AI's business model includes offering pilot trials to customers first, then based on feedback, Guardforce AI develops custom solutions for the client. In 2022, Guardforce AI quickly expanded its business footprint, and deployed robots for front desk reception, indoor disinfection and delivery, and hotel self-check-in. It's currently partnering with third-party providers to develop robots that can offer travel information and booking services to tourists.

One of the challenges that Guardforce AI encountered during the development process is to enhance interactions between humans and robots in various languages. Lin Jia, President of Guardforce AI, says that at first, its development team collaborated with local service providers to realize features like voice recognition and text-speech conversion, but it became increasingly challenging when the company's business expanded into more countries. Adopting different software solutions in different regions means that Guardforce AI needed to change its robotics technological framework each time, which was time- and effort-consuming. At the same time, it also had to pay extra attention to ensure that its services are in compliance with the data privacy regulations in each country.

To streamline its development work and support parts of the RaaS services, Guardforce AI would need to adopt certain services from a global cloud platform that is compliant across regions. After

evaluating different offerings, the company found that Google Cloud (<https://cloud.google.com/>) not only supports highly accurate voice recognition in English, but also enables smooth human-robot interactions in less spoken languages. As a result, it eventually adopted Google Cloud.

"To bring our service robots to various regions around the world more efficiently, we need to deploy parts of our technological framework in a cloud environment that is compliant with data regulations across countries. On top of meeting this need, Google Cloud also supports highly natural human-robot interactions in multiple languages.' explains Lin.

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Quickly delivering robotic solutions to global users with Speech-to-Text and Text-to- Speech AI

Guardforce AI's main manufacturer partner is in China. Therefore, Guardforce AI adopted Chinese providers' artificial intelligence (AI) speech services in its initial phase. However, other than Mandarin, the AI speech services provided by Guardforce AI's Chinese provider in other languages did not deliver sufficient performance. When the company entered

markets like Thailand and Malaysia, it needed to collaborate with local service providers to rebuild its solutions. At that time, Guardforce AI hadn't established an overarching framework that allows third-party developers to build custom solutions independently. Therefore, when launching services in a new region, the team needed to go through the whole development process from design to testing, which required frequent and long communication with local service providers.

Now, Guardforce AI deploys a unified technological framework for its voice services in Google Cloud and [Dialogflow](https://cloud.google.com/dialogflow) (<https://cloud.google.com/dialogflow>) to enable conversations between users and robots, [Speech-to-Text](https://cloud.google.com/speech-to-text) (<https://cloud.google.com/speech-to-text>) and [Text-to-Speech AI](https://cloud.google.com/text-to-speech) (<https://cloud.google.com/text-to-speech>) to convert between speech and text. In the process of the framework construction, Google Cloud partner [Master Concept](https://hkmci.com/about-master-concept/) (<https://hkmci.com/about-master-concept/>) helped Guardforce AI evaluate various solutions to find the most effective way to enhance the accuracy and efficiency of its natural language processing technology.

The unified framework saves enormous time and effort: Guardforce AI can launch its solutions without searching for multiple third-party providers for each language, which has helped save approximately 60 percent of labor costs. On top of that, the AI speech services that Guardforce AI used before were tightly coupled with their corresponding hardware. In comparison, the Google Cloud tools

offer higher flexibility and usability by not being coupled with any hardware. Guardforce AI's development team can therefore focus on software improvement, without the need to understand the manufacturing process of certain robot hardware.

Adopting Google Cloud has also helped Guardforce AI enhance its service quality. According to its internal tests, Speech-to-Text not only supports highly accurate speech recognition in English, but also recognizes speech in less spoken languages like Thai and Malay accurately enough to enable natural human-robot interactions, which meets the needs of Guardforce AI's customers in its primary markets. Besides, the hyper realistic speech generated by Text-to-Speech AI has improved the interaction performance of Guardforce AI's service robots.

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Shortening robotic solution development cycle with Compute Engine

Guardforce AI leverages Compute Engine (<https://cloud.google.com/compute>) to run robot model training, simulation, and complex tasks. With the autoscaling feature of Compute Engine, the Guardforce AI team is able to meet the computing demand increase without manually adding machines. It can also choose the CPU and GPU specifications that correspond to its actual needs. As a result, the reduction in operational cost varies between 20% to 40% depending on projects. When designing new functions and new architectures, applying Google's recommended model helped Guardforce AI to consider responding to failures and

how to use the various services provided by Google to respond to failures, which shortened its product development cycle by half.

Other Google Cloud tools have also facilitated the development of more diverse solutions. Guardforce AI offers three plans for its robotic solutions.

Besides one-off purchase and rental, its customers can also choose to put robots in places where people regularly pass by and receive revenues from ads displayed on the screens of robots. To realize effective ad targeting, Guardforce AI is developing an ad targeting solution based on information collected from robots like environment, information, locations, and Q&A interactions.

The new solution requires transforming information collected from robots into semantics and outputting ad display recommendations through several algorithms. Leveraging the globally distributed computing resources of Google Cloud (such as Compute Engine, Cloud Storage, and Cloud CDN) and Google Maps Platform

(<https://developers.google.com/maps>) for identifying robotic locations, Guardforce AI is able to reach shorter service response time and show targeted ads more quickly.

Ensuring data privacy compliance around the world

Following the enactment of the General Data Protection Regulation (GDPR) in the EU and similar laws in other countries, it has become a prerequisite for tech companies to process user data securely

and legally when providing data-related services. For example, for Guardforce AI to provide a service robot that works as a hotel front desk agent or assists in travel booking, it has to collect users' personal identity information while processing the data in compliance with local regulations. Lin points out that for a solution provider, it is extremely challenging to achieve full compliance in all countries around the world on its own. With all the data compliance certifications acquired by Google Cloud, Guardforce AI can easily ensure full data compliance of its products in various markets.

"Our robotic business runs in several different countries in Asia Pacific and the United States. It would have required a lot of time and energy if we had to look into data regulations in every country on our own. Google Cloud helps us realize data compliance more efficiently by providing cloud products in compliance with data privacy regulations across regions," he adds.

Expanding use scenarios of service robots with large language models

After successfully running selected RaaS services with Google Cloud, Guardforce AI plans to expand the use scenarios of its service robots by further improving human-robot interactions through the latest large language models like [Gemini](https://gemini.google.com/) (<https://gemini.google.com/>). Building a large language model is very resource-demanding, so for a solution provider like Guardforce AI, leveraging the most

advanced AI technology of Google Cloud can greatly increase its product development speed.

Lin Jia says, "The industry-leading AI technology of Google Cloud has enabled us to enhance the performance of our robotic solutions. Moving forward, we will leverage more deep learning technologies of Google Cloud to train our robot models, generate more accurate responses and feedback, and reach our business goal of realizing human-robot collaboration in the cloud in a better way."

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